1. Problem analysis and advice

Status: working draft for team/client validation Project: James - RVL IHM lab-report workflow
Evidence date: repository notes through 2 November 2026

1.1 Organization and user need

RVL performs inspections and prepares hazardous-material and IHM documentation. The project materials describe a workflow that moves from project setup and field registration through laboratory analysis, quality review and reporting. Lab results must be associated with the correct sample and source in the inspected object. The final assessment remains the responsibility of an authorised inspector.

The business need recorded in the project context is to reduce repeated manual handling, avoid omissions, preserve traceability, and reuse structured object history. The initial use case is deliberately narrow: help prepare IHM Part I results from registered samples and a lab report.

Evidence: docs/context.md summarises the RVL onboarding and work process. Verify this understanding with the client during the demo and record any corrections.

1.2 Current and proposed process

Set up job and register field findings

Current/manual work described in project notes: Create project and record sources/samples, locations and evidence
James prototype contribution: Project and sample registration screens are described in the app and build notes

Receive laboratory report

Current/manual work described in project notes: Read report and identify result rows
James prototype contribution: Paste report text or upload PDF; digital text is extracted directly, scanned pages can use local OCR when installed

Match results to samples

Current/manual work described in project notes: Compare report rows with field register
James prototype contribution: Rules propose clear links; uncertain eligible cases can be passed to the AI step; conflicts abstain

Check and approve

Current/manual work described in project notes: Inspector checks findings and corrects errors
James prototype contribution: Named inspector approves, corrects, rejects or leaves items undecided

Prepare deliverable

Current/manual work described in project notes: Compile report and preserve supporting records
James prototype contribution: Draft IHM sections and ODT export from approved data; evidence and decision history are retained

The process description is based on project notes, not a fresh time-and-motion study. Manual baseline time and current error rates have not yet been measured.

1.3 Problem statement

Preparing an IHM report requires people to transfer and reconcile information across field records, sample registers, lab reports and a report template. This creates repetitive work and opportunities for omissions or incorrect links. James explores whether structured registration, assisted extraction, conservative matching and human review can make this work more traceable and efficient.

The project has not yet demonstrated a measured reduction in processing time or reliable extraction

across real laboratory layouts. Those are validation questions, not established benefits.

1.4 Scope

In scope for the prototype/pilot: one IHM/HazMat Part I workflow; fictional or explicitly approved test data; project and sample registration; report text/PDF intake; extraction and sample-link proposals; completeness warnings; human review; draft report output; audit/evidence capture; repeatable evaluation.

Out of scope for the current prototype: autonomous approval; unrestricted report-layout support; broad survey types; unapproved real client data or AI services; production deployment and operational support.

1.5 Options and provisional multi-criteria analysis

The following comparison is a team working assessment, not client-approved scoring. Scores are 1 (weak) to 5 (strong). For risk and cost, a higher score means lower/manageable risk or cost. Criteria and weights should be agreed with the client before presenting this as a final recommendation.

Potential process value

Weight: 25%
A. Manual process with templates: 2
B. Digital workflow + deterministic rules: 3
C. Digital workflow + rules and AI assistance: 5

Safety and controllability

Weight: 25%
A. Manual process with templates: 5
B. Digital workflow + deterministic rules: 5
C. Digital workflow + rules and AI assistance: 3

Feasibility within three months

Weight: 20%
A. Manual process with templates: 5
B. Digital workflow + deterministic rules: 4
C. Digital workflow + rules and AI assistance: 3

Traceability

Weight: 15%
A. Manual process with templates: 2
B. Digital workflow + deterministic rules: 4
C. Digital workflow + rules and AI assistance: 5

Cost, privacy and dependency manageability

Weight: 15%
A. Manual process with templates: 5
B. Digital workflow + deterministic rules: 4
C. Digital workflow + rules and AI assistance: 2

Weighted score / 5

Weight: 100%
A. Manual process with templates: 3.80
B. Digital workflow + deterministic rules: 4.00
C. Digital workflow + rules and AI assistance: 3.65

How to read this: option B scores highest as a near-term controlled pilot because it improves structure and traceability without depending on an unapproved model. Option C has greater potential for difficult cases, but its value and risk are not yet evidenced; the current AI is a

replay/stand-in, not a live service. This does not mean AI should be abandoned. It means real AI should remain a gated experiment and must beat the rules baseline on agreed test data without increasing silent wrong links.

Recommendation: continue with the digital workflow and rules as the dependable baseline. Trial AI linking only after RVL approves the tool and data conditions, and only on uncertain cases with strict validation and inspector review. Treat cloud hosting as a separate later decision after a pilot security and operating model is agreed.

Confirm with stakeholders: criteria and weights; what counts as acceptable error; approved data and AI provider; whether there is a second reviewer; pilot hosting requirements; measured manual baseline.

1.6 Advice and decision gates

1. Confirm workflow, report fields, sample-code conventions and sign-off roles with RVL.
2. Agree synthetic or approved representative documents and have an IHM expert label the expected fields and links.
3. Validate extraction and rule-linking first. Keep errors visible and route ambiguity to people.
4. Evaluate real AI against the same held-out cases only after tool and data approval.
5. Consider a limited hosted pilot only after privacy, access, backup, retention and support requirements are defined.